

RFID-Controlled Solenoid Access Circuit Package

Prototype-oriented design for a 12 V cabinet lock that unlocks for a timed interval after an authorized RFID card is scanned.

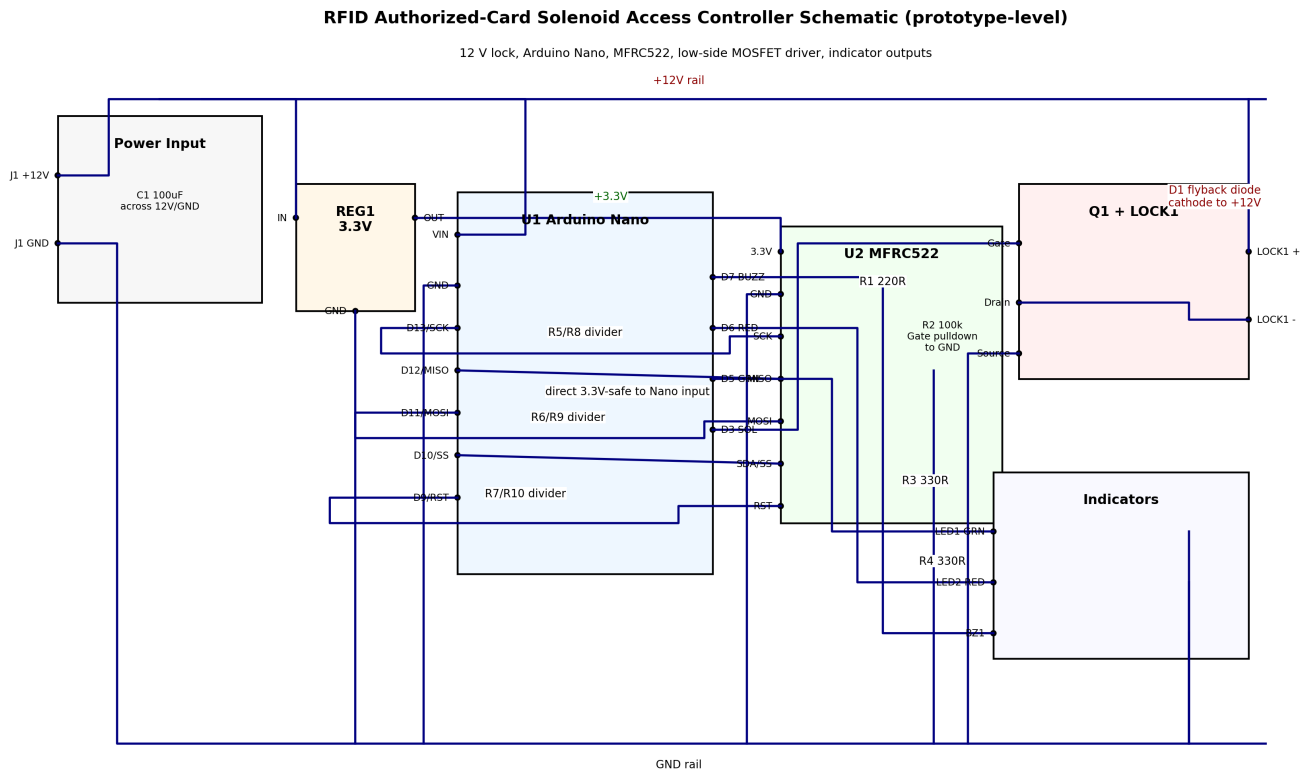
Summary

This design uses an Arduino Nano and an MFRC522 RFID reader. The Nano compares scanned card UIDs against an internal allowlist. When a match is found, the Nano drives a logic-level MOSFET that energizes a 12 V cabinet-lock solenoid for about 3 seconds. A flyback diode protects the driver stage, and LEDs plus an optional buzzer provide user feedback.

Main assumptions

- 12 V DC cabinet lock solenoid, intermittent-duty, about 0.4-0.8 A.
- Arduino Nano powered from the 12 V source via VIN for simplicity.
- MFRC522 powered from a dedicated 3.3 V regulator.
- Simple resistor-divider level reduction on Nano to MFRC522 logic lines for prototype use.
- Authorized card UIDs stored in firmware source code.

Schematic



QA note: this is a clear functional wiring/schematic-style drawing intended for review and prototyping. It is not a fabrication CAD file.

Bill of materials

Item	Refs	Qty	Description	Unit \$	Ext \$	Supplier
1	U1	1	Arduino Nano	27.1	27.1	Arduino Store
2	U2	1	MFRC522 RFID reader module	9.9	9.9	RobotShop
3	Q1	1	N-channel MOSFET	2.5	2.5	Infineon / estimate
4	D1	1	Flyback diode	0.54	0.54	DigiKey
5	REG1	1	3.3 V regulator module	3.15	3.15	Maker Store
6	LOCK1	1	12 V cabinet lock solenoid	8.99	8.99	Estimated market price
7	R1	1	Gate resistor	0.1	0.1	Estimate
8	R2	1	Gate pull-down resistor	0.1	0.1	Estimate
9	R3,R4	2	LED resistors	0.1	0.2	Estimate
10	R5,R6,R7	3	SPI level-shift resistors	0.1	0.3	Estimate
11	R8,R9,R10	3	SPI divider resistors	0.1	0.3	Estimate
12	LED1	1	Green LED	0.2	0.2	Estimate
13	LED2	1	Red LED	0.2	0.2	Estimate
14	BZ1	1	5 V active buzzer	1.0	1.0	Estimate
15	C1	1	Bulk capacitor 12 V rail	0.35	0.35	Estimate
16	C2,C3	2	Decoupling capacitors	0.08	0.16	Estimate
17	J1	1	DC barrel jack	0.75	0.75	Estimate
18	J2,J3	2	2-pin screw terminals	0.45	0.9	Estimate

Total estimated BOM cost: \$56.74 USD

Pricing note: official/store pricing was used where easily available; several commodity passives, connectors, and the MOSFET were estimated to keep the BOM numerically complete.

Firmware deliverable

Arduino sketch file: `/workspace/rfid_solenoid_access/firmware/rfid_solenoid_access.ino`

The sketch initializes the MFRC522, checks card UIDs against an allowlist, energizes the solenoid output for a configurable interval, and drives green/red status indicators plus an optional buzzer.

Important limitations and next steps

- For production access control, use secure credential storage instead of plain UID-only authorization.
- If the lock current is high or cable length is long, consider a dedicated lock supply and stronger transient suppression.
- Many RC522 modules are strictly 3.3 V devices; full logic-level translation is preferable in a hardened design.
- A door position switch, tamper detection, and fail-safe/fail-secure analysis should be added before deployment.